

VaLoRe

ML-powered real estate valuation platform

Milestone 3

INFO9023 - Machine Learning System Design

Antoine DECKERS (s2170999)

Hoang Linh BUI (s2503303)

Duy Vu DINH (s2401627)

May 11th, 2026



Overview - What is VaLoRe?

- ML real estate valuation platform aims to automatically estimate the market value of properties using historical sales data, property characteristics, and contextual information.

Value Creation:

- For buyers: fair property price → avoid overpaying
- For sellers: set competitive listing prices → faster transactions
- For real estate agents: support data-driven pricing decisions
- For banks / investors: assist in property risk and valuation analysis



Milestone 3 Deliverable

- **Streamlit** dashboard contains:
 - **online data serving**: user requests are sent in real time to a deployed **FastAPI** prediction service
 - Dynamic EDA: the dashboard automatically retrieves data analysis from **BigQuery**
 - Model performance monitoring
- This approach was chosen over offline serving to enable scalable real-time inference and live cloud integration.

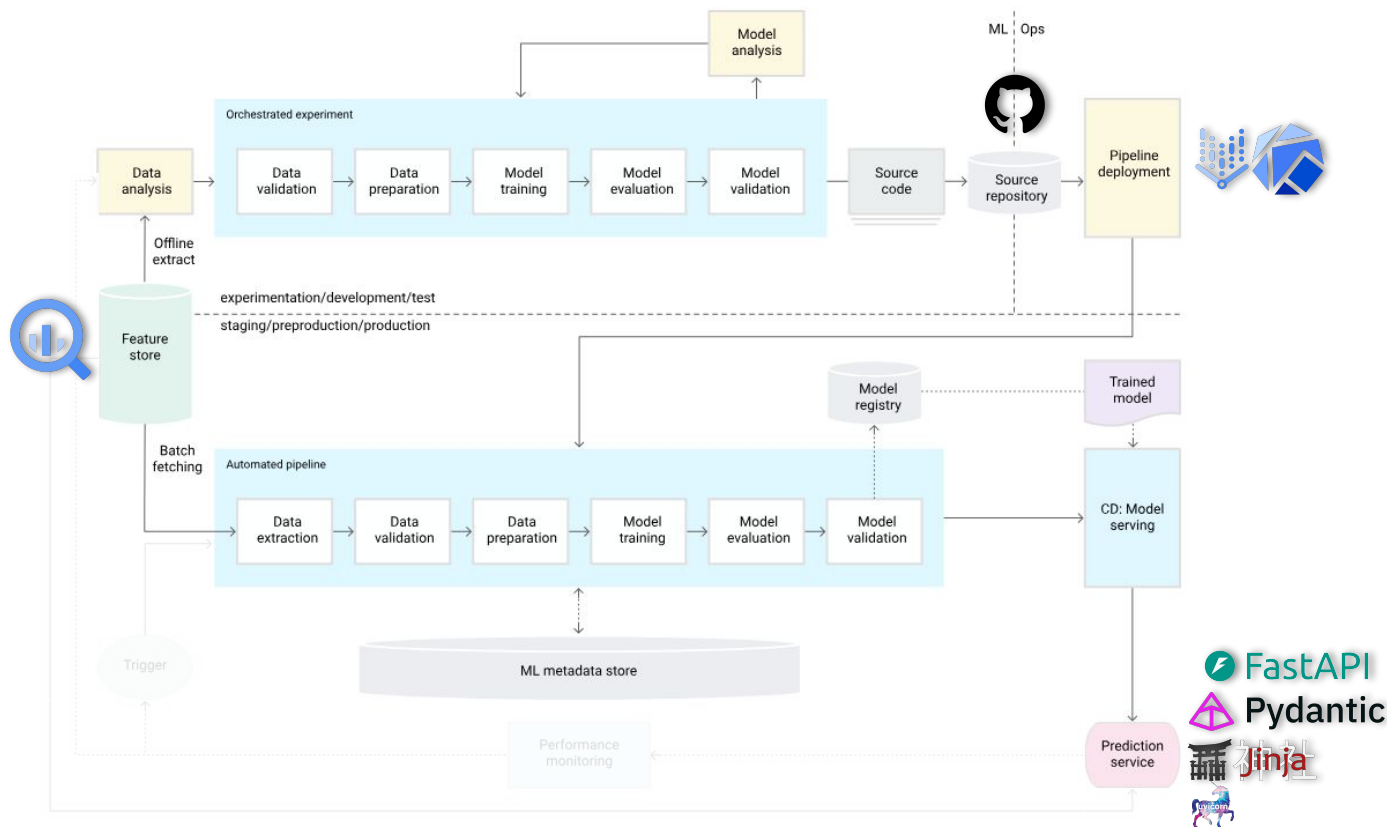


Feature description

Structural	sqm_living	Total square area of the interior living space
	sqm_lot	Total square area of the land parcel
	sqm_above	Interior living space located above ground level
	sqm_basement	Interior area located in the basement
Quality & View	waterfront (0-1)	Binary indicator for properties with a waterfront view or access
	view (0-5)	Qualitative rating of the property's surrounding scenery
	condition (1-4)	Evaluation of the house's current physical state and maintenance
	grade (1-13)	Construction and architectural quality
Neighborhood	sqm_living(lot)15	Average living (parcel) area of the 15 nearby homes



Project Overview | MLOps Pipeline



Dashboard - **DEMO TIME**

- <https://valore-dashboard-826612837630.europe-west1.run.app/>
- Cloud run settings :

Container concurrency Limited to 10 concurrent users per container to maintain stable Streamlit performance	CPU / RAM allocation Each container allocated 2 vCPU and 2 GiB RAM for responsive real-time processing
Scaling strategy Automatic saling to max 20 containers for high availability	Load testing (Locust) Validated stability under 50 concurrent users using Locust (cfr github / last slides)

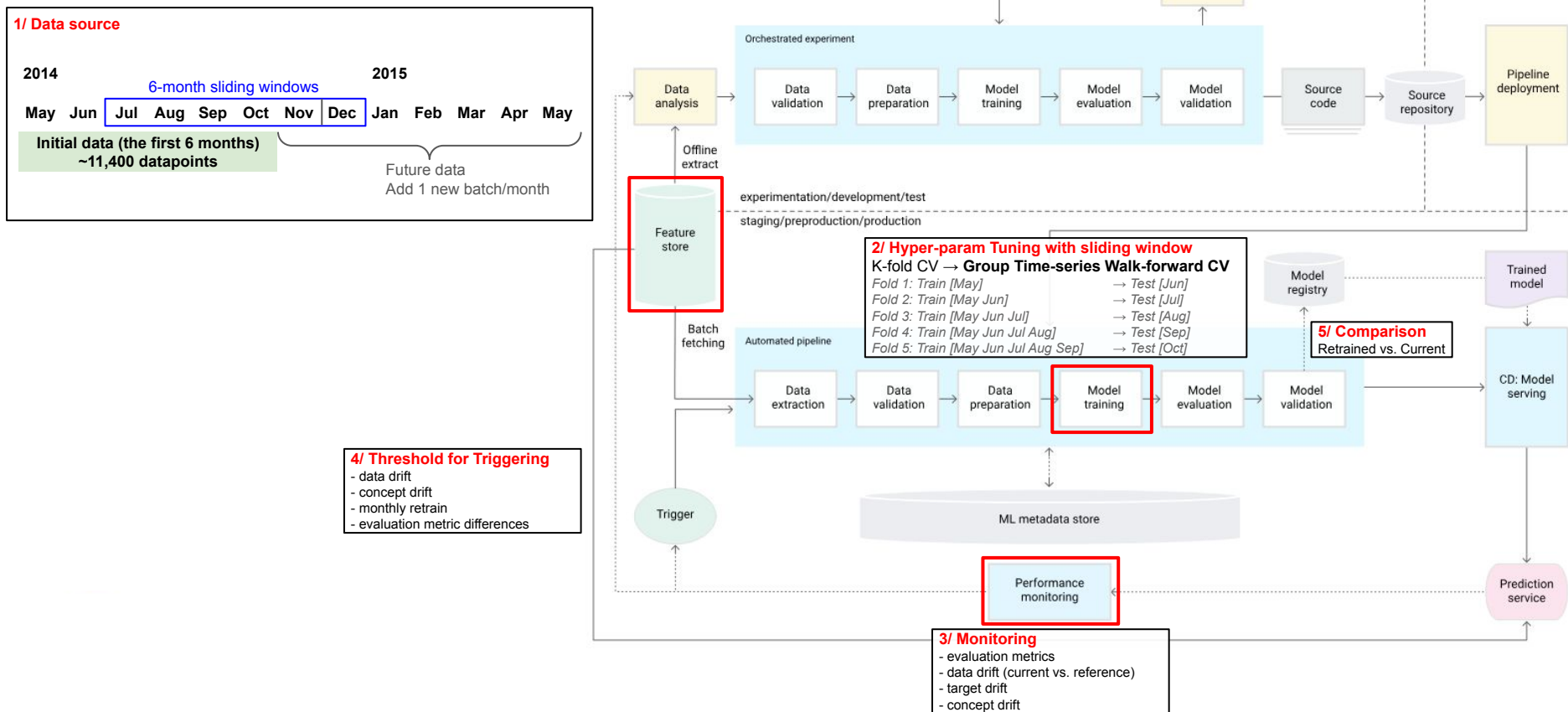


Limitations

- Data is “frozen”
- No retrain trigger
- No monitoring
- No model validation



Future Work | Complete MLOps time-series pipeline



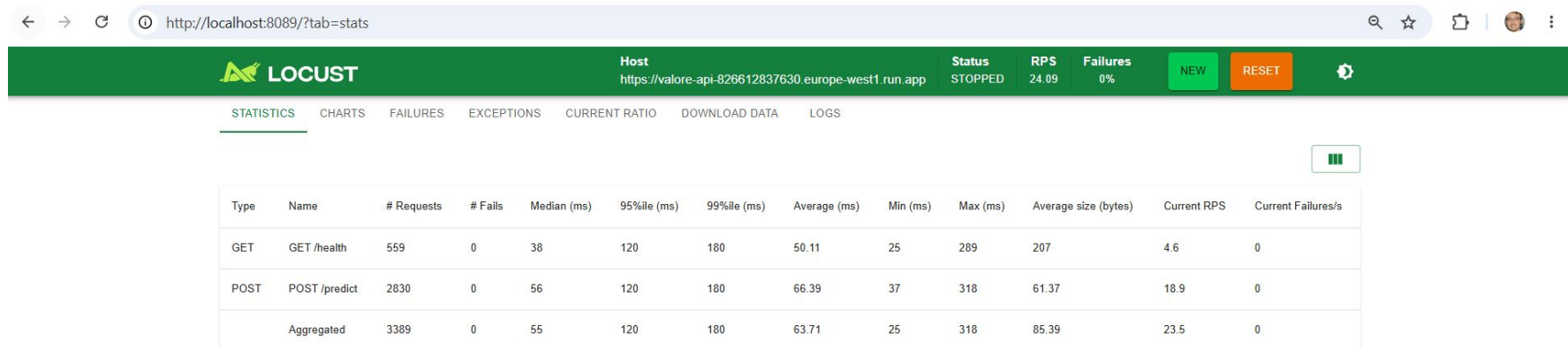
Thank you for your listening !

Q&A

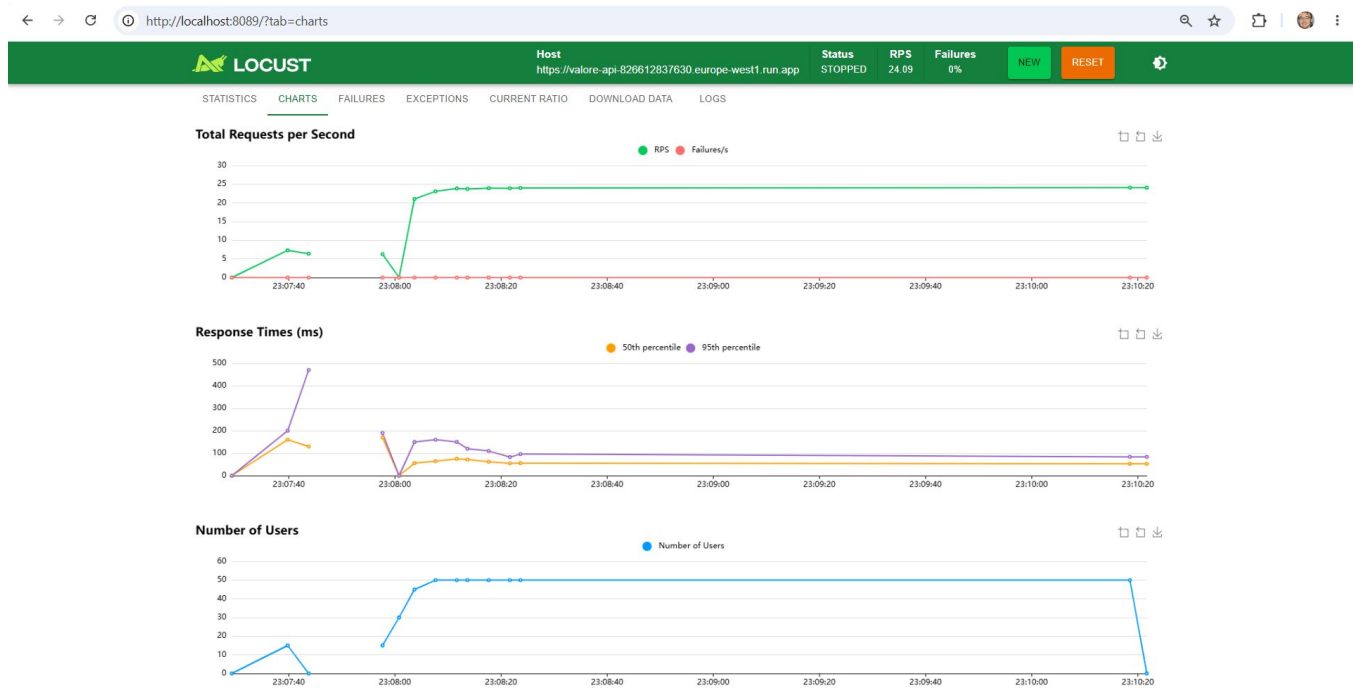


Locust api

<https://valore-api-826612837630.europe-west1.run.app>



Locust api

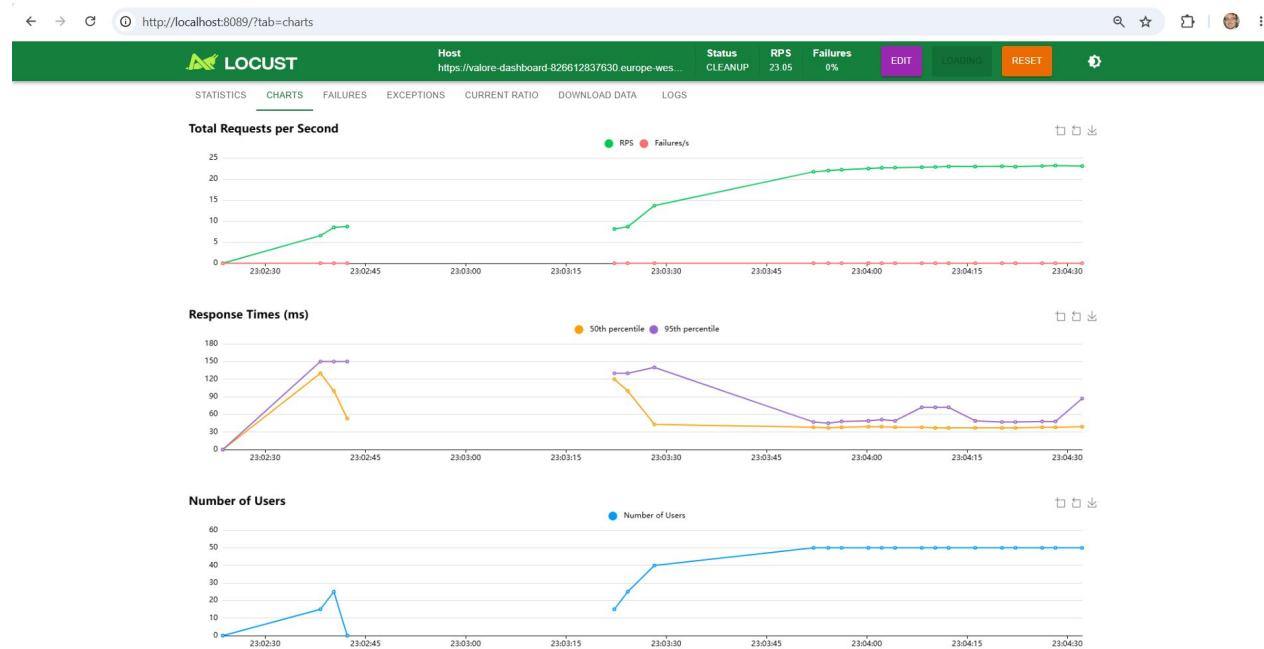


ABOUT

Locust dashboard



Locust dashboard



ABOUT